Run report: run-20261022T0844Z (deep 2-volcano, prime conductor gap)

Descriptive record of the measurement run. The results report is `../../README.md` its readings are not repeated as claims here.

What ran

item	value
protocol	`PROTOCOL.md`, frozen in commit `f7b4d94` before any measurement; erratum (190, not 191, curves per side) written as the sessions were starting, before any result was read
build	`run_pipeline.sh build run 52 6 16384 20261022 2 6 4 2` (construct, two `volcano-rho --ecbench` exports, certify, spec)
run	`run_pipeline.sh run run`: three `ecbench run --cpus none --allow-busy` sessions, concurrently, one lock each
start / end (UTC)	sessions started 08:45:19 (BSGS), 08:45:43 (A/A), 08:47:14 (main); ended 08:51:20, 08:57:11, 09:20:39 (2026-10-22)
host	Intel Xeon Processor @ 2.10 GHz, 4 vCPU shared cloud VM, Linux; isolation L0; no PMU
ecbench binary	sha256 `a59c4014ff52100ed10c06b5d27a0572f9474d24c137fd18b24798237024f81f`
sessions' git	`f7b4d94`, `git_dirty: true` (untracked run directory)

Sessions

session	arm (method)	workloads	runs planned / recorded / verified	failures
main	`rho-neg` (`rho.negation`)	760 curve models × 6 targets	4560 / 4560 / 4560	0
A/A	`rho-neg-aa` (`rho.negation`, seed ⊕ 0xAA00000000)	380 × 4	1520 / 1520 / 1520	0
BSGS	`bsgs-neg` (`bsgs.negation`)	380 × 2	760 / 760 / 760	0

Session and spec ids:

- main: session `ECBS1h41a98e7c2141`, spec `ECS1h7d9934575ec8` (`run/specs/spec.json`)
- A/A: session `ECBS1h4034bfcbccc4`, spec `ECS1h0d34427ad5a6` (`spec-aa.json`)
- BSGS: session `ECBS1h658495a369ec`, spec `ECS1ha55be72ab53f` (`spec-bsgs.json`)
- **Seeds.** Measurement seed 20261022. Target law `planted_shared` (`uniform_scalar_sha256_shared_v1`), target seed 20261022.
- **Settings.** Warmup 0, rounds 1, order shuffle, timeout 600 s; no run timed out.

Audit

`ecbench verify --replay 24 --exit-code` on each session:

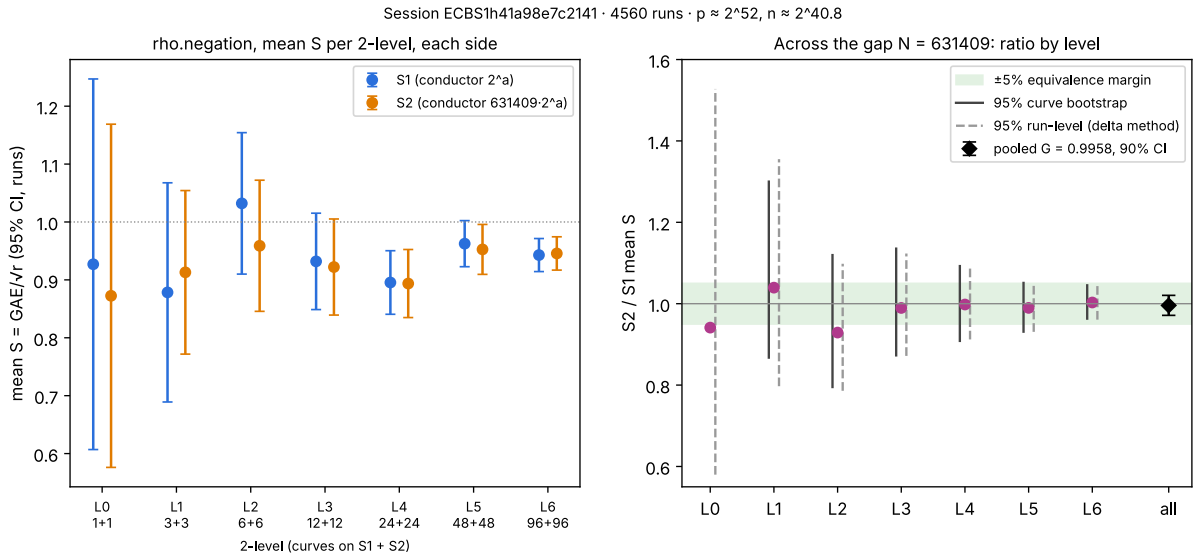
session	result
spec	`audit OK: 4560 records, 4560 verified, 24 replays reproduced, 0 problems`
spec-aa	`audit OK: 1520 records, 1520 verified, 24 replays reproduced, 0 problems`
spec-bsgs	`audit OK: 760 records, 760 verified, 24 replays reproduced, 0 problems`

The receipts are `../../run/sessions/*.audit.json`.

Recorded values (measured; computed by `volcano-stats`, `../../analysis.json`)

quantity	S1	S2
`rho-neg` runs	2280	2280
`rho-neg` mean S	0.94288 ± 0.0203	0.93889 ± 0.0208
per-curve mean solve wall time, min–max (ms, both sides)	224.3–571.3	(pooled)
per-curve mean solve wall time, mean (ms, both sides)	394.2	(pooled)

The intervals are 95% over runs.



Raw records

- `../../run/sessions/{spec,spec-aa,spec-bsgs}/records.jsonl.gz` (uncompressed bytes pinned by `../../run/sessions/MANIFEST.sha256`)
- `session.json`, `host.json`, `spec.json` and `plan.json.gz` in each session directory
- the sidecars `../../run/export-s{1,2}/curves.jsonl`
- the class and isogeny `../../run/pair.json`
- the side certificates `../../run/certificates.json`